

SOP-RE-225: Rotating Equipment Vibration Monitoring and Fitness-for-Service

Document owner: Rotating Equipment Department **Effective date:** 2024-07-01 **Review cycle:** 24 months

Section 1. Purpose and Scope

Governs continuous vibration monitoring of critical rotating equipment (compressors, pumps, turbines) and the fitness-for-service review triggered by sustained trend increases.

Section 2. Alarm Thresholds

Clause 2.1 — Each critical rotating machine has a vibration alarm setpoint established from OEM allowable limits at commissioning, recorded on its Rotating Equipment Datasheet.

Section 3. Sustained Trend Review

Clause 6.3 — Sustained vibration trend increases on critical rotating equipment exceeding the Clause 6.5 alarm-persistence threshold require a formal bearing fitness-for-service review before continued operation is authorized.

Clause 6.5 — Alarm-persistence threshold: three consecutive monitoring periods above the general alarm setpoint, or any single reading above 150% of setpoint, whichever occurs first.

Section 4. Fitness-for-Service Outcome

Clause 6.4 — A fitness-for-service review may establish a revised sustained-operation limit specific to the evaluated failure mode (e.g., a thrust-bearing limit distinct from the general alarm setpoint), which supersedes the general setpoint for that unit going forward. Any sustained reading above the revised limit for more than one monitoring period is treated as a fitness-for-service exceedance requiring load reduction and re-evaluation before return to full rated speed.

Section 5. Records

Clause 8.1 — Vibration trend data and fitness-for-service reviews are filed to the Rotating Equipment Integrity File. Retain for the operating life of the unit plus 10 years.